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Recession *Probability* Engine

A machine-learning approach to U.S. business-cycle forecasting

AUTHOR
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PERIOD
1950 – present

VALIDATION
11-fold walk-forward, out of sample

STACK
XGBoost · SHAP · FRED

0.948 *AUC*

ROC-AUC, PUBLISHED OUT-OF-SAMPLE SERIES 1971–2025

11.3 *months*

AVERAGE ADVANCE WARNING

0 *months*

ABOVE THE ALARM LINE WITH NO RECESSION AHEAD, 1985–2025

6 / 6

OUT-OF-SAMPLE RECESSIONS CAUGHT

ABSTRACT

We present a machine-learning system that produces a monthly probability estimate of a U.S. recession occurring within the next 12 months. The architecture addresses three structural challenges of macro forecasting: a sparse positive-class history (~11 NBER-defined recessions since 1950), retroactive label definition, and the severe credibility cost of false alarms. A single XGBoost model is trained on public FRED data and validated by strict walk-forward — one test period per recession, every label used only once its 12-month outcome is known — and the published reading is smoothed so that an alarm needs sustained evidence. Version 4 traced and fixed the walk-forward's worst flaw: a 21-month false alarm in 1984–86 that appeared when v2.1 stopped each test model from training on labels it could not yet know. The fix was chosen by a test whose selection rule was fixed before it ran. On the published out-of-sample series — each month scored by the model that would have been live, 1971–2025 — the system catches all six recessions with an average 11.3 months of warning, reaches ROC-AUC 0.948 and PR-AUC 0.813, and has not once been above the alarm line without a recession ahead since November 1984: zero such months from 1985 to 2025, and 10 over the whole record, down from 46. Run through the same walk-forward, the literature's standard yield-curve model (Wright, 2006) scores ROC-AUC 0.840 with 28 false-alarm months, 21 of them in 2023–24, although it warns earlier before 2001 and 2008. This public edition reports the main results and what each part of the method guarantees; the exact rules and settings are in the full technical paper, available from the author on request.

WHAT'S NEW IN V4.0

The public edition

This edition reports version 4 as it stands: its main results, unchanged, and what each part of the method guarantees — without its exact rules and settings, which are in the full technical paper, available from the author on request.

The 1985 fix

Version 4 removes the 1984–86 false alarm the v2.1 label rule exposed (21 months above 50%, peak 82%): false-alarm months fall from 46 to 10, none of them after 1984, and every recession is still caught (§4.6).

A pre-registered choice

The fix's one free setting was chosen by a rule written down and committed before the test ran, on five random seeds, and every candidate is reported in the full technical paper.

Metrics on the published series

Results are measured on exactly the series that is charted and published, with PR-AUC, sensitivity, specificity and the Brier score alongside ROC-AUC, recessions caught, lead time and false-alarm months (§5.1).

A classic benchmark

Wright's (2006) yield-curve probit, run through the identical walk-forward: ROC-AUC 0.840 against 0.948 and 28 false-alarm months against 10 — but with longer leads before 2001 and 2008 (§5.5).

01 INTRODUCTION

The hardest easy problem in applied ML.

Recession forecasting occupies an unusual position in applied machine learning. The prediction target is rare — roughly 11% of post-war months fall within an NBER-defined recession window — its definition is retroactive, and the asymmetric cost of errors is severe. A false alarm is not merely an incorrect prediction; it is a credibility-destroying event that may cause the next genuine signal to be ignored.

Three structural challenges compound the difficulty:

01 **Label scarcity.**
The United States has experienced approximately 11 NBER-defined recessions since 1950 — fewer than 130 positive-class samples across 75 years of monthly data. A dataset size that would be considered trivially small in most ML contexts.

02 **Retroactive definition.**
The NBER typically announces recession start dates six to twelve months after the fact. Recent training labels are therefore provisional and subject to revision; models trained on the most current data are learning from unresolved ground truth.

03 **Forward contamination.**
A 12-month forward label means a sample from early 2019 carries a recession label of 1 simply because COVID arrived in Q1 2020 — not because of any cyclically learnable pattern in 2019. Without an explicit architectural response, the model learns spurious associations.

> *The objective was not to build the most complex forecasting system, but a credible one — one that catches every major recession with meaningful advance warning, while maintaining a false-alarm rate low enough to preserve operational utility over time.*
>
> — PROJECT THESIS

02 DATA & FEATURES

From public FRED data to fifty-seven engineered features.

All data are sourced from FRED, the St. Louis Federal Reserve's public macroeconomic database: a few dozen indicators of the U.S. economy and its financial markets, aligned to monthly frequency.

FEATURES

Raw series are never used directly as model inputs. Each indicator is turned into a few features that describe its level and how fast it is changing. This puts heterogeneous series on comparable scales and brings out the rate-of-change signals that lead the underlying levels.

Where a series begins after 1950 — VIX before 1990, for instance — the corresponding features are left as *proper missing values*. No imputation is applied. The production model handles structural missingness natively, learning to weight features relative to their actual historical availability.

CLASS IMBALANCE

With roughly 11% recession-month prevalence, the dataset is severely imbalanced. Synthetic oversampling (SMOTE) is explicitly rejected: generating artificial recession months would corrupt the temporal structure of the data and produce training observations that violate chronological ordering. Instead, the rare recession months are weighted more heavily in training, rebalancing the loss without fabricating history. One consequence matters for reading the output: the weighting shifts probabilities upward by design, so the signal is best read as a risk score on the scale its 50% threshold was set on, not as a calibrated frequency (§5.1).

03 ARCHITECTURE

One model, chosen from ten.

Several alternative feature sets and ten algorithms were compared through the same walk-forward before the production configuration was fixed.

WHY XGBOOST

Ten algorithms were evaluated — XGBoost, HistGradientBoosting, Logistic Regression, Random Forest, Gradient Boosting, AdaBoost, Extra Trees, SVC, MLP, Naive Bayes — on AUC across walk-forward folds and false-positive rate at the calibrated threshold. XGBoost dominated on both

criteria across all feature sets and time periods. It was selected as the sole production model for three reasons:

- Native handling of structural missingness without imputation.
- Performance – It was the only model to consistently generate robust and reliable predictions across all evaluation windows.
- SHAP decomposability allows feature-level attribution at every monthly inference.

Through v3 the pipeline still trained a HistGradientBoosting model beside it and blended the two at weights 1 : 0, so the second model never moved a number. v4 removes the blend: production is one XGBoost model, with every published value bit-identical to the blended version. HistGradientBoosting remains available as a comparison algorithm.

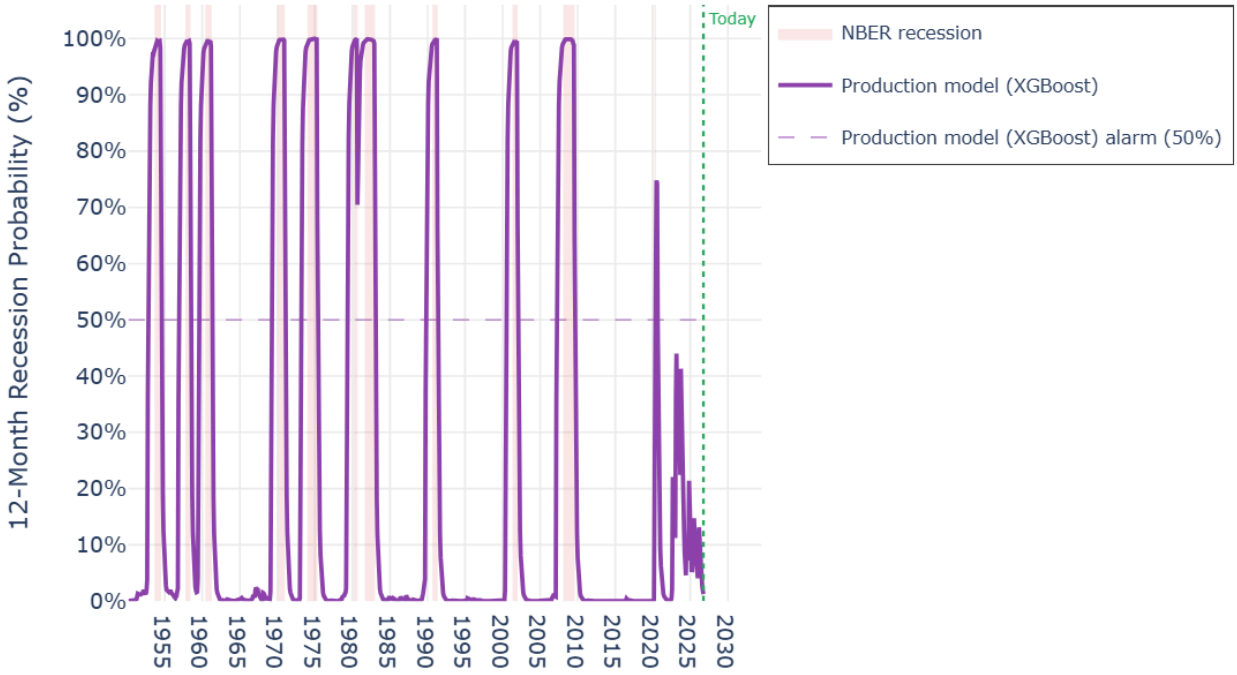


FIG. 1
FULL HISTORY

Production-model recession probability, 1950 – present, as published (smoothed). NBER recession windows shaded in pink. The model tracks every post-war recession and remains suppressed during the 2011 European debt crisis, the 2015–16 China slowdown, and the 2022–23 rate-hike cycle.

Note: before 2019 this curve is the model's in-sample fit to events it was trained on. Its training data ends in January 2019 (§4.2), so every month plotted after that is genuine out-of-sample prediction. Out-of-sample performance across the whole record is in Fig. 2–4 and §5.

04 METHODOLOGY

What keeps the test honest.

4.1 — WALK-FORWARD VALIDATION

Standard k-fold cross-validation is inappropriate for time-series macro data: training on post-2005 data to predict pre-2000 outcomes leaks future distributional information into the past. Walk-forward validation enforces a strict chronological constraint — each fold trains on all data up to a fixed cutoff and evaluates on all subsequent data. A cutoff follows each historical recession, producing 11 folds covering the full record. Each fold accumulates one additional recession episode in its training data, simulating the progressive accumulation of evidence that occurs in real-time deployment.

Label resolution: no row before its outcome is known (v2.1).

A training row dated m carries a label built from months m+1 … m+12, so it can only be used once those twelve months have passed. Version 2.1 enforced that rule at every test boundary, which removed a genuine look-ahead. It looked nearly free — average mature-fold AUC 0.959 → 0.955 — and the average hid a 21-month false alarm in 1984–86, which version 4 fixed (§4.6).

4.2 — STRUCTURAL SAFEGUARDS

A fixed training window.

The live model's training data end in **January 2019**, and the window moves only at defined points, never month by month. Recent labels are provisional — NBER revises its recession dates retroactively, and a row dated m asserts an outcome over months m+1 … m+12 that has not resolved for rows near today — so the training window never takes in labels that are not yet settled.

It also keeps deployment and validation identical. The production model is trained on exactly the rows of the last walk-forward test model: **the model that is deployed is the model that was validated.** And because the production model never observes a month after January 2019, the entire post-2019 segment of the full-history chart is genuine out-of-sample prediction.

COVID excluded.

The pandemic months are removed from the training data. Samples from early 2019 carry a recession label of 1 because COVID struck in Q1 2020, not because of any learnable cyclical deterioration, and the 2020 recession was an exogenous biological shock. Retaining the period would teach the model to associate ordinary cyclical weakness with pandemic-scale anomalies. The same months are excluded from false-positive counting and from every metric in §5.

Publication-lag alignment (v2).

FRED series are stamped with their *reference period*, not their release date: March CPI carries a March 31 timestamp but is not published until mid-April. v1 indexed every series at its reference-period end, which silently exposed the model to data before it would have been observable in real-time deployment. v2 places every series where it would actually have been published. This eliminates a category of look-ahead leakage that was particularly acute for the quarterly indicators the model relies on for the early stages of a contraction. It moved AUC by only +0.008; it was done anyway, because a model that cheats with time is worthless even when it cheats only a little.

Strict prediction anchor (v2).

FRED quarterly series can extend the feature index past the current month — corporate profits, for instance, often publishes a forward-dated month-end stamp. v1's anchor logic only skipped the exact current month, which silently allowed future-dated rows to surface as the "current" inference under specific calendar conditions. v2 enforces a strict filter to the last complete month strictly before today, eliminating an entire class of forward-bias that could quietly contaminate the production probability.

4.3 — A SIGNAL THAT NEEDS SUSTAINED EVIDENCE

Raw probability output from any macro model contains noise. A single month above the alarm threshold does not constitute a reliable signal; isolated spikes are common during mid-cycle slowdowns and rate-hike episodes that do not resolve into recessions. Since v2 the published reading smooths the model's monthly output, so that an alarm needs sustained evidence.

The smoothing is *path-aware*: a gradual build-up over several months counts for more than a single-month spike from a calm base. A sustained high signal crosses the alarm threshold sooner than a fixed-window confirmed filter would allow; an isolated spike, however large, fades before it can.

> **WHY IT REPLACED THE CONFIRMED FILTER**
>
> v1 used a confirmed-signal filter requiring several consecutive months above threshold before activating. It was robust but had two structural weaknesses: it was **path-blind** (a gradual build-up was treated identically to an isolated spike preceded by calm months), and it imposed a **fixed activation delay** regardless of signal strength.
>
> The smoothing resolves both. In testing on mature folds it preserves the low false-alarm profile while recovering the 2001 Dot-Com recession that the confirmed filter missed entirely — the latter's required streak kept breaking against the noisier 2001 build-up.

The smoothing directly affects the AUC reported in walk-forward validation: AUC is computed on the smoothed, published series, not the raw model output. Smoothing reduces month-to-month noise and produces a more stable ranking of pre-recession months against normal months, which raises AUC. This is intended — the smoothed series is the operational signal the

system produces — but it means the AUC should be compared with other smoothed-signal AUCs, not raw-probability benchmarks. The benchmark in §5.5 goes through the same smoothing.

Hysteresis emerges *implicitly* from the smoothing: a brief dip during a sustained high period cannot instantly reset the alarm, because the published reading reflects the recent months, not only the latest one. No explicit deactivation threshold is required.

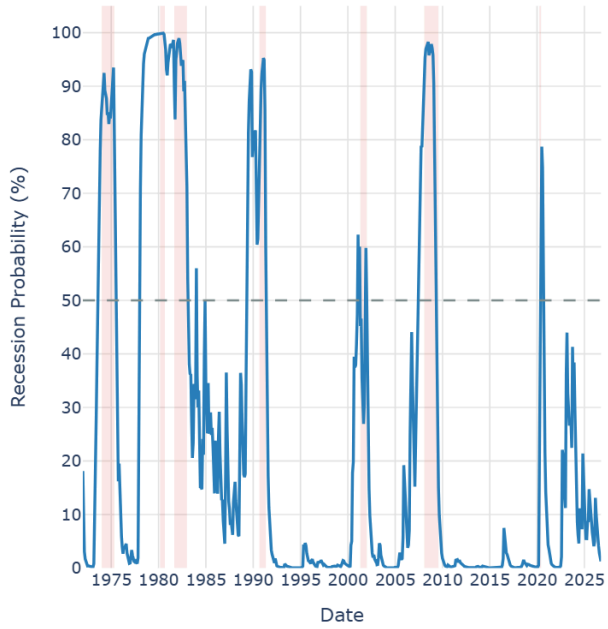


FIG. 2 PUBLISHED SERIES

Walk-forward predictions, folds 4–11, as published: each month predicted by the test model that would have been live, then smoothed — the series scored in §5. Short exceedances are suppressed; structural alarm periods (1973, 1980, 1981, 1990, 2001, 2007) remain. The 2001 build-up crosses threshold three months before the recession begins.

4.4 — THRESHOLD CALIBRATION

A model outputs a probability; an alarm requires a threshold. The production threshold is fixed at **50%** — selected by sweeping candidate levels over the OOS predictions from mature folds for the value that minimized dangerous false-positive streaks (sustained above-threshold readings not followed by an NBER recession within 12 months) while preserving maximum recession coverage. The natural center of the probability scale satisfied both criteria, with a clean interpretation: the alarm fires when the model has genuine, balanced-evidence conviction rather than marginal elevation.

4.5 — FEATURE VALIDATION

Before the production feature set was fixed, every candidate feature had to earn its place in walk-forward tests. **Most did not: they degraded AUC, increased the false-positive rate, or both.** The dominant failure mode was multicollinearity: many candidates were overlapping

proxies for the same underlying phenomenon, diluting signal without providing orthogonal information.

4.6 — FIXING 1985: A PRE-REGISTERED TEST

The v2.1 label rule was correct, and it also introduced the walk-forward's worst flaw. Before it, the 1984–88 out-of-sample signal peaked at 42.9%. Under it — same data, same model — the signal climbed to 81.8% and stayed above 50% for 21 consecutive months, from November 1984 to July 1986. The 2001 and 2008 warnings were unchanged.

The cause lies in the test model that covers the mid-1980s. Its training data ended with the 1981–82 recession: of the 45 months before that point, 44 are labeled “recession ahead”, with the Fed funds rate averaging about 13.6%. It had no example of high rates in a healthy economy, so “high rates mean recession” was a rule with no exceptions in its training data. SHAP attributes about half of the 1984–86 rise to the Fed funds level alone (+1.04 of +1.92 log-odds), with CPI and oil next.

Version 4 changes how the test models are trained. The change has one free setting, which could easily be tuned until 1985 disappears, so it was chosen under a rule written down and committed before the test ran: every recession caught on every one of five random seeds, then the fewest false-alarm months, then the longest average lead, then PR-AUC, with near-ties going to the more conservative choice. Every candidate is reported in the full technical paper.

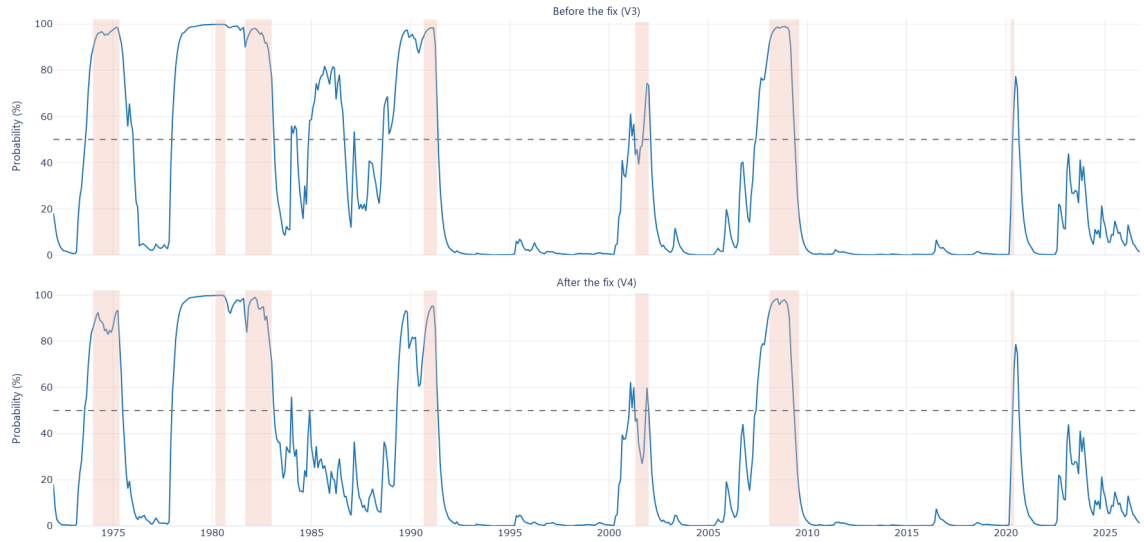


FIG. 3
THE 1985 FIX

Walk-forward published signal, folds 4–11, before the fix (top, v3) and after it (bottom, v4), production seed. The 1984–86 plateau is gone; what remains are single months in December 1983 (56%) and November 1984 (50.05%). Everything from 2021 on is identical.

What remains of 1983–84 is two isolated months: December 1983 at 56%, and November 1984 at 50.05%, after the Fed had pushed rates back toward 11% and the curve had gone nearly flat. Both fall back the following month (§7.3). The cost is small and measured. Averaged over the

five seeds, the v3 rule's 11.6-month lead and 83% sensitivity at 50% become 11.4 months and 80% (Table 1 shows the production seed); on one seed in five the 2008 warning shrinks to 2 months (it stays at 8 on the production seed). None of it touches the live model: its training data are unchanged (§4.2), and every reading since 2021 is identical.

05 RESULTS

Out-of-sample performance, measured on what is published.

Through v3 the headline numbers averaged each fold's metrics over every later decade — fold 4, trained on data up to the early 1970s, was scored on 2023. That measures how a model ages, not what was published. v4 headlines the stitched series instead: each month is scored by the fold model that would have been live at the time, exactly as Fig. 2 draws it, from 1971 to the last month with a resolved label (October 2025). The pandemic months are excluded throughout. A recession counts as caught if the published signal crosses 50% in the 18 months before its onset; a false-alarm month is any month above 50% outside every window running from 18 months before an onset to 3 months after the recession ends.

5.1 — THE PUBLISHED SERIES

TABLE 1 — PUBLISHED OUT-OF-SAMPLE SERIES · PRODUCTION SEED

METRIC	V3 RULE	V4	WRIGHT (2006)
Recessions caught	6 / 6	6 / 6	6 / 6
Lead, months (1973 · 1980 · 1981 · 1990 · 2001 · 2008)	4 · 18 · 18 · 18 · 3 · 8	5 · 18 · 18 · 16 · 3 · 8	8 · 17 · 18 · 18 · 8 · 14
Average lead, months	11.5	11.3	13.8
False-alarm months, 1971–2025	46	10	28
False-alarm months since 1985	**26**	**0**	**22**
ROC-AUC	0.934	0.948	0.840
PR-AUC	0.801	0.813	0.747
Sensitivity at 50%	81.9%	81.2%	68.8%
Specificity at 50%	85.7%	94.0%	90.2%
Brier score	0.101	0.072	0.124

The v3-rule and v4 columns are the same model, data and seed; only the training rule differs. Leads are capped at 18 months by the detection window. Eight of v4's ten false-alarm months are December 1977 – July 1978, the 1980 warning arriving 26 months before onset; the other two are the 1983–84 months of §4.6. From January

1985 onward the v4 series has none at all, through four recessions, the 2022–23 stress test and the live period to date; the v3 rule has 26 in 1985–89 and the benchmark 22, in 1989 and 2023–24. The benchmark is described in §5.5.

ROC-AUC asks whether pre-recession months are ranked above calm ones; PR-AUC asks the same of the rare class and is the stricter of the two when positives are scarce (Davis & Goadrich, 2006). Sensitivity and specificity are measured at the 50% alarm line. The Brier score, the mean squared distance between the probability and the outcome, improves the most (0.101 → 0.072), but it should be read with care: because the model is class-weighted (§2), its output is a risk score on the threshold's scale rather than a calibrated frequency.

5.2 — FOLD BY FOLD: HOW A MODEL AGES

The per-fold view, reported as the headline through v3, scores each fold model on *all* later data — far beyond the segment in which it would ever be used. It remains a useful stress test of how a model ages.

TABLE 2 — PER-FOLD RESULTS · EACH FOLD SCORED ON ALL LATER DATA · PUBLISHED SIGNAL

FOLD	RECESSION ENDED	RECESSIONS IN TRAINING	AUC	FP / YR
4	Nov 1970	4	0.940	0.2
5	Mar 1975	5	0.905	0.3
6	Jul 1980	6	0.957	0.1
7	**Nov 1982**	**7**	**0.977**	**0.0**
8	Mar 1991	8	0.981	0.0
9	Nov 2001	9	0.990	0.0
10	Jun 2009	10	n/a †	0.1
11	Apr 2020	11	n/a †	0.0
Average (mature folds 4–9)			**0.958**	**0.1**

† No completed out-of-sample recession is eligible for AUC scoring after 2009 (COVID-2020 is excluded as exogenous). On identical data the v3 rule averages 0.954; fold 7, the 1985 culprit, moves from 0.943 and 0.2 FP/yr to 0.977 and 0.0. Across folds 8–11 the months above 50% outside any recession window total 6 — in 2006, early 2023, and one month after 2001 — down from 15. Average advance warning in this view, excluding COVID: ***13.2 months****.*

5.3 — THE 2001 RECESSION: CAUGHT, WITH A SHORT LEAD

In v1, the test model for 2001 (trained on data up to the early 1990s) assigned only ~35% to the 2001 Dot-Com recession and the confirmed filter never activated. From v2 on — same model, same training data, smoothed signal — the signal crosses threshold in January 2001, **3 months before** the recession begins. The 2001 build-up arrived through several months of mid-50s–

mid-60s readings, repeatedly interrupted by lower months that broke the confirmed filter's consecutive-month streak. The smoothed signal keeps that build-up instead of resetting on each dip, recovering a weak-but-genuine signal that v1's filter discarded.

The lead time is shorter than at other detections in the modern record for reasons explained in §7.1: the most diagnostic features for a corporate-led, equity-and-CapEx contraction had observed at most one prior recession by the time that test model was trained. The model could not produce a strong signal, but the signal it did produce is no longer thrown away.

5.4 — THE 2022–23 STRESS TEST

The 2022–23 period was the most demanding false-positive challenge in the modern dataset. The Federal Reserve raised rates by 525 basis points in 16 months; the 10Y–2Y yield curve inverted to its deepest level since 1981; and surveyed economists put the odds of a recession above 60%. Most quantitative macro models — including the univariate yield-curve heuristic — generated sustained false-positive signals.

The production signal rose to 44% in February 2023 but never crossed the 50% threshold. The SHAP decomposition of that month shows why, and corrects v3, which attributed the rise to the Fed funds level. The inverted curve (+0.46) and the pace of hikes over the past year (+0.42) did push up, but the largest upward pushes came from falling corporate profits (−3% year over year, +0.75), depressed consumer sentiment (+0.57) and rising involuntary part-time work (+0.55). The Fed funds *level* pushed down (−0.60): at 4.3% it is ordinary by the standards of the 1954–2019 training record (§7.4). Low jobless claims (−0.42), a Sahm reading of zero (−0.36) and positive industrial production (−0.25) held the line. In the per-fold view, two older fold models — fold 8, trained through 1991, and fold 10, through 2009 — brush above 50% for one or two months in early 2023 (peaks near 58%); the fold-11 model and the production model stay below throughout.

5.5 — AGAINST A CLASSIC BENCHMARK

The natural question about a 57-feature model is whether it beats a simple one. The literature's standard simple model builds on the yield-spread probits of Estrella and Mishkin (1998). Wright (2006) shows that adding the level of the nominal Fed funds rate to the 3-month / 10-year term spread improves both in-sample fit and out-of-sample accuracy, and that the strongest recession signature is "an inverted yield curve with a high level of the nominal funds rate." We ran that two-variable probit through the identical walk-forward — the same test periods and training rules, pandemic exclusion, smoothing, 50% line and metrics, with its inputs aligned to release dates exactly as in the production pipeline. Only the model and its inputs differ, so any difference in Table 1 comes from the model, not the test.

The benchmark catches all six recessions too, and warns earlier: 8 months before 2001 against 3, and 14 before 2008 against 8, because the yield curve inverted well ahead of both. It pays for that with 28 false-alarm months — 21 of them consecutive, from April 2023 to December 2024, the inversion that did not become a recession — and with lower accuracy on every month-level measure: ROC-AUC 0.840, PR-AUC 0.747, sensitivity 69%. The multidimensional model gives up a few months of lead on two episodes for a far quieter signal, and in 2023–24 the quiet signal was the right one.

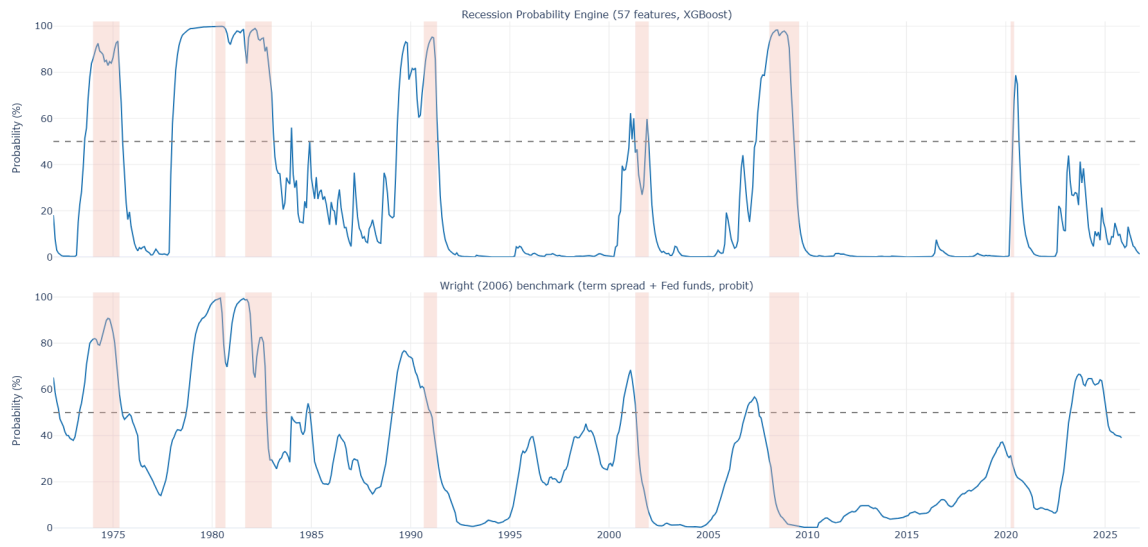


FIG. 4 BENCHMARK

Published signal, folds 4–11, for the production model (top) and Wright's two-variable probit (bottom), both run through the same walk-forward and smoothing. The benchmark's smoother curve rises earlier before 2001 and 2008 but stays above 50% from April 2023 to December 2024.

Versus the univariate inversion rule.

The 10Y–2Y Treasury yield-curve inversion is the most widely cited single-variable recession predictor. A naive rule — "alarm when the spread turns negative" — has historically preceded U.S. recessions with reasonable consistency. In 2022–23, however, it generated a sustained false alarm exceeding 26 months. Over the same period, this model's production signal did not activate.

> *The structural explanation is not that the yield curve is wrong, but that it is context-free. A curve inversion in a full-employment economy with accelerating industrial production is fundamentally different from an inversion accompanied by rising unemployment and tightening credit. A multidimensional model can represent that distinction; a univariate threshold cannot.*

Versus published results.

Comparisons across papers are rough — different samples, frequencies, horizons and evaluation designs — so they are offered as orientation, not a ranking. Wright reports out-of-sample root mean square errors of 0.33–0.40 for recursive recession-probability forecasts over 1980–2005, and 0.34 for his best four-quarter models. Scored on the same window, the v4 series has an RMSE of 0.25 (0.23 when averaged to quarters) and a ROC-AUC of 0.967, against an RMSE of 0.33 under the v3 rule. Yazdani (2026), revisiting machine-learning recession models on 1959–2024 data, finds tree ensembles strongest, with out-of-sample AUC above 0.9 — the range in which this system sits.

06 EXPLAINABILITY

Every probability comes with its decomposition.

Because the system operates as an early-warning tool, explaining *why* the probability is elevated is as operationally important as the probability itself. Every monthly inference is accompanied by a SHAP (SHapley Additive exPlanations) decomposition across three analysis layers, and each month's full decomposition is now also stored as data, not only as a chart.

6.1 — GLOBAL FEATURE IMPORTANCE

Ranked by mean absolute SHAP value, the most influential features are the federal funds level, building permits and corporate profits, followed by the yield-curve slope, initial jobless claims and housing starts. This hierarchy was not encoded manually — the model learned it from the data alone. It is consistent with established macro theory (the cost of capital leads; the real economy confirms), which is reassuring, though not by itself proof of causal structure.

6.2 — HISTORICAL TIMELINE

Feature contributions plotted as time series across the full historical record reveal the recession-specific signature of each episode. The 2008 recession shows housing starts and permits contributing sharply from 2006–07, well before the employment data broke. The 1980 and 1981 recessions show the federal funds rate and monetary tightening as the dominant driver. The 2001 episode shows corporate profits and an inverted curve activating while consumer indicators remained relatively contained — the distinctive fingerprint of an equity and capital-expenditure bust.

6.3 — MONTHLY WATERFALL

For any given month, the SHAP waterfall chart shows the signed contribution of each feature to that month's probability — how much each indicator pushed the estimate above or below the historical base rate. This transforms the black-box probability into an interpretable macroeconomic summary: a decision-maker can identify exactly which sectors are driving the current risk assessment.

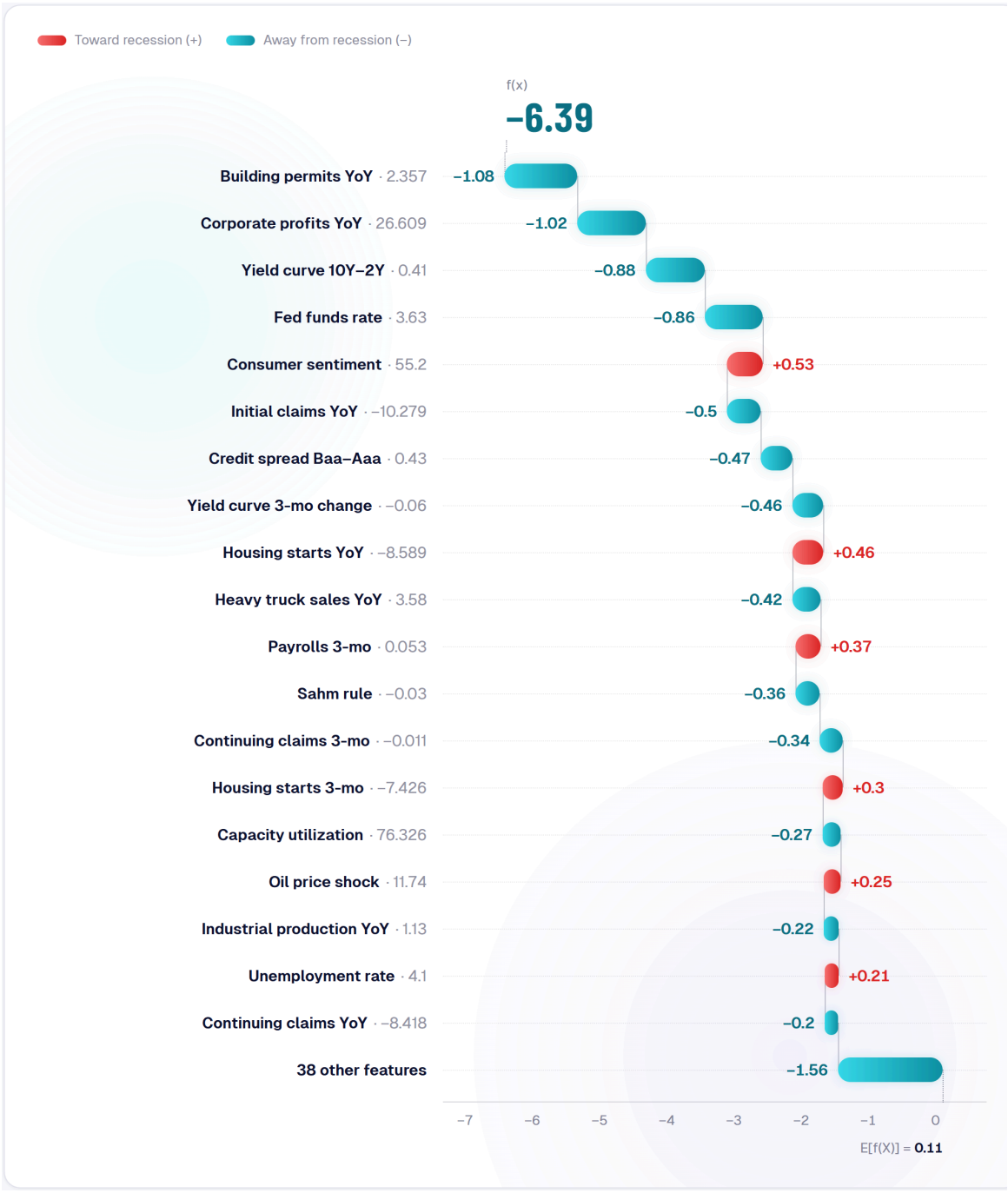


FIG. 5
DRIVERS

The drivers chart as published on Kleriox for August 2026: each bar is one indicator's push toward (red) or away from (teal) a recession, applied in sequence from the model's average output, E[f(X)], to the month's output, f(x); the 19 largest are shown and the other 38 as one total. The reading is held down by building permits, corporate profits, the yield curve and the Fed funds rate; low consumer sentiment is the largest push up.

6.4 — FINGERPRINTS OF PAST RECESSIONS: A REGIME SHIFT AROUND 2000

Running the waterfall on individual pre-recession months reveals the distinctive macroeconomic signature of each episode — and a structural break in those signatures around the year 2000. Pre-2000 recessions are dominated by classic monetary and capacity-tightening

features: the 1989 decomposition is led by the Fed funds rate, the 1973 episode by capacity utilization and business-loan growth at the peak of an overheating cycle. From 2000 onward, that classic signature recedes; corporate profits take over as the single dominant driver.

The model learned this regime shift directly from the data. It is consistent with the macro literature: the post-Volcker monetary regime stabilized inflation, lowered the relative volatility of policy rates, and shifted business-cycle dynamics toward credit and corporate-balance-sheet channels. All panels below are computed with the current production model; the four historical months lie inside its training data, so they show what it *learned*.

PRE-2000 · CAPACITY- AND FED-FUNDS-DRIVEN

September 1989

F(X) = +4.97

TOP DRIVER — FED FUNDS RATE · +1.43

Eleven months before the 1990 recession. The Fed funds rate at 8.99% leads, with initial jobless claims (+1.38) close behind and corporate profits (+0.92) confirming. A monetary-policy story.

February 1973

F(X) = +4.51

TOP DRIVER — CAPACITY UTILIZATION · +1.54

Ten months before the 1973–75 recession. Capacity utilization at 87.8% — a peak-of-cycle overheating reading — leads, followed by business-loan acceleration (+1.02) and the pace of rate hikes (+0.71). A pre-Volcker overheating signature where capacity and credit lead the monetary rate.

POST-2000 · CREDIT & CORPORATE-BALANCE-SHEET DRIVEN

February 2007

F(X) = +4.79

TOP DRIVER — CORPORATE PROFITS · +1.07

Eleven months before the Great Recession. Corporate profits, jobless claims (+1.02), housing permits (+0.72) and housing starts (+0.68) push positive in concert — the housing-and-credit GFC signature. The Fed funds rate at 5.25% pulls the probability *down* (−0.39).

May 2000

F(X) = +3.54

TOP DRIVER — CORPORATE PROFITS · +1.22

The first recession of the new regime. Eleven months before the Dot-Com contraction, corporate profits, an inverted curve (+1.13) and business-loan growth (+0.82) lead; the Fed funds rate at 6.02% contributes only +0.12 — a fraction of its 1989 weight.

> *Read across the four panels: the dominant feature shifts from capacity and the funds rate (1973, 1989) to corporate profits (2000, 2007). The model is not just picking different signals — it is picking the same hierarchy that the macro regime itself produced.*

WHAT THE MODEL SEES

The four panels above show what the model *learned* from past recessions. The panel below is different: it is a *prediction* for a month after the production model's training data ends (January 2019). It shows what the model was signalling, not what it absorbed in training.

March 2025 F(X) = -2.85

TOP SUPPRESSOR — CORPORATE PROFITS · −1.09

A live prediction during a period of macro stress that *did not* become a recession. Corporate profits, business-loan growth (−0.77 and −0.72 on its three-month and annual changes) and the Fed funds level (−0.69) push the probability down; falling consumer sentiment pushes up (+0.87 on its annual change, +0.58 on its level) but is outweighed. The published reading never approaches the threshold.

07 LIMITATIONS

What the system cannot do — by design.

7.1 — The 2001 walk-forward anomaly: caught, but barely

The system detects the 2001 Dot-Com recession with 3 months of lead — substantially shorter than at other modern episodes, and the underlying signal never climbs as high as it does at other recessions. The previously-reported v1 “miss” was, mechanically, the confirmed filter discarding a weak-but-genuine signal whose month-to-month variability kept resetting its consecutive-month counter; the smoothed signal captures it without requiring uninterrupted threshold exceedance.

The structural model-level argument behind the weakness remains valid. The 2001 recession was a localized corporate equity bust, not a broad consumer or housing contraction. The most diagnostic indicators for that type of event — corporate credit, bank lending and business-investment data — begin only in the 1990s, and had recorded at most one recession episode by the time the model that tested 2001 was trained. A model that has never observed a “CapEx bust while consumers are healthy” scenario in its training history cannot be expected to produce a strong signal for it. The simple yield-curve benchmark warns five months earlier here (§5.5) — at the price of a two-year false alarm in 2023–24.

7.2 — The cost of a fixed training window: no post-2019 learning

The fixed training window buys deployment/validation identity at a real price, and the price grows every month: the production model's last training observation is January 2019 — over seven years before the current inference — and the window moves only at defined points.

The model has therefore never seen 2021–2024 — not the post-pandemic inflation surge, not the fastest tightening cycle since Volcker, and, most relevant to reading its current output, not the 2022–23 episode in which a deep and sustained yield-curve inversion, the single most-cited recession indicator, resolved without a recession. A near-miss is the most informative negative in the record, and the production model is trained on none of the recent ones.

The honest statement of the tradeoff: the system is being asked to recognize a 2026 recession using a feature-to-outcome mapping estimated through January 2019, in exchange for a model whose behavior is exactly what §5 reports. The window will move forward over time.

7.3 — Two months left in 1983–84

The 1985 fix removes the 1984–86 false alarm, but not every crossing: December 1983 (56%) and November 1984 (50.05%) remain. The second followed the Fed's push back toward 11% and a nearly flat curve — close to the configuration Wright identifies as the strongest recession signature. Both are single months. Removing them would mean either giving the model hindsight or adding a persistence rule that delays every genuine warning by a month, so they are disclosed instead.

7.4 — The Fed funds level is read against the training era

The Fed funds rate is the model's most important feature (§6.1), and its meaning depends on the rates a model has seen. The test model trained only through the Volcker years read 8% as danger in 1985; the production model, trained on 1954–2019, reads 4–5% as ordinary, and in 2022–23 the level was the second-largest force holding the signal down (§5.4). Dropping the feature removes the 1985 problem in the walk-forward, but in our tests it also made the live model sit above 50% for every month of 2023. We keep it — Wright (2006) finds the nominal level informative beyond the spread — and flag the dependency: an expansion with persistently higher rates than the training record would be a regime the model has not seen.

7.5 — Scope

This system produces a single output: the probability that an NBER-defined recession will occur within the next 12 months. It does not forecast recession depth, duration, sector composition, equity-market performance, or GDP growth rates. It is an early-warning signal, not a macroeconomic model.

7.6 — Feature-selection ceiling

The production feature set was fixed by the search described in §4.5, but that search is itself a heuristic. A more exhaustive search, or a principled selection algorithm with stronger statistical guarantees, would almost certainly surface a feature configuration with better out-of-sample behavior than the one shipped here. At the current stage of the work I was not able to achieve those better results; the production set is the best configuration recovered under the available compute and search budget, not a proven optimum. The same caution applies to every setting chosen by looking at the walk-forward — threshold, filter, feature set: with six out-of-sample recessions, a backtest can reward luck. The 1985 fix was chosen under a pre-committed rule for exactly that reason.

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08 CONCLUSION

Architectural honesty over algorithmic complexity.

This project demonstrates that a disciplined, temporally honest machine-learning pipeline can produce a credible and operationally useful early-warning signal for U.S. business cycles. The key contributions are architectural rather than algorithmic:

- Walk-forward validation that mirrors deployment chronology: every label is used only once its outcome has resolved — on final-vintage FRED data by design, with revisions treated as measurement noise.
- Explicit structural mechanisms against label leakage, retroactive revision, exogenous contamination and publication-lag look-ahead, plus a strict prediction anchor on the production inference.
- A smoothed, path-aware signal that converts raw monthly probabilities into operational alarms — sensitive to gradual build-ups, robust against isolated spikes.
- Decisions tested in the open: the 1985 fix was chosen by a test whose selection rule was committed before it ran, and the model is compared with the literature's standard benchmark through the identical walk-forward.
- Full SHAP explainability connecting each monthly probability to its macroeconomic drivers, stored month by month as data.

The resulting system catches all six countable recessions in its out-of-sample record with an average 11.3 months of warning, ranks months with ROC-AUC 0.948 (PR-AUC 0.813), and has not been above the alarm line without a recession ahead since November 1984 — 10 such months over the whole record, down from 46 under the v3 rule, against 28 for a classic yield-curve

model run through the same test. It withstood the most demanding false-positive environment in recent decades — the 2022–23 rate-hike cycle — without the production signal activating. It does this not by being more complex than the yield-curve heuristic, but by being more *complete*: contextualizing any single signal within the full macroeconomic environment, and translating raw monthly probabilities into operational alarms with a path-aware filter that respects both the integrity of a build-up and the noise of a spike.

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A ABOUT THIS EDITION

What this edition leaves out.

This public edition describes what each part of the method guarantees, not how it is built. The main results are all here; the exact rules and settings, and the supporting experiments, are in the full technical paper, available from the author on request.

COLOPHON

Set in Source Serif 4 and Inter.
Labels in JetBrains Mono.

Claude, by Anthropic, was used to help with this project.

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